

Microsoft Corporation

MSFT: NASDAQ

Intrinsic Value Analysis - Discounted Cash Flow Model

GGM Implied Price (Primary Method)	Exit Multiple (Cross-Check)	Market Price (Valuation Date)	Verdict (Primary)
\$385	\$574	\$497	OVERVALUED

Analyst	Manuel Carvalhinha Alves Costa
Valuation Date	30 June 2025
Analysis Date	May 2026
Model	Integrated 5-Year DCF · 13 Tabs
WACC	8.12% · Terminal Growth: 3.0% · Exit Multiple: 20×

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Valuation Conclusion

Primary Method (GGM): Intrinsic value of \$385/share - a 23% discount to the June 2025 market price of \$497. On this basis, Microsoft is OVERVALUED at the valuation date. The Exit Multiple method (\$574) is presented as a cross-check only, anchoring to current market pricing of mature cash flows rather than long-run fundamental value.

Core Thesis

The thesis is a timing bet, not a quality bet. Microsoft's business quality is not in question - EBIT margins have expanded from 41.6% in FY2021 to 45.6% in FY2025, and the company generates over \$136B in annual operating cash flow. The valuation question is whether the market correctly prices the duration and magnitude of the current CapEx suppression cycle.

The base case models CapEx normalisation from 22% of revenue in FY2026 to 12.5% by FY2030, consistent with Microsoft's pre-AI historical profile (FY2022: 12.0%, FY2023: 13.3%). This drives FCFF expansion from \$83B to \$172B - a 107% increase over five years and the mechanical engine of the bull thesis for long-term holders.

Valuation Summary

Method	Implied Price (USD)	vs. Market (\$497)	Role
Gordon Growth Model	\$385	-23%	Primary method
Exit Multiple (20× EV/EBITDA)	\$574	+15%	Cross-check only

Scenario Analysis

Input	Bear	Base	Bull
Revenue CAGR FY26-30	~9%	~11%	~13%
EBIT Margin FY30	44%	46%	48%
CapEx % Revenue FY30	22%	12.5%	11%
Terminal Growth Rate (g)	2.0%	3.0%	3.5%
Exit EV/EBITDA Multiple	16×	20×	24×
GGM Implied Price	\$222	\$385	\$494
Exit Multiple Implied	\$411	\$574	\$764
Verdict (GGM)	Overvalued	Overvalued	Fairly Valued

The bear case (-55% GGM downside) requires CapEx to remain elevated at 22% through FY2030. The bull case (-1% GGM downside) requires CapEx to normalise to 11% alongside accelerating revenue growth - plausible but optimistic.

Sensitivity - GGM Implied Price (USD)

WACC \ g	2.0%	2.5%	3.0% ★	3.5%	4.0%
7.12%	399	436	482	542	619
7.62%	362	392	429	474	532
8.12% ★	331	356	385	421	466
8.62%	305	325	350	379	414
9.12%	282	300	320	344	372

1. Terminal growth rate is the dominant variable - a 200bps move in g at base-case WACC produces a \$135/share range (\$331-\$466).
2. WACC is the second most sensitive input - a 200bps increase at base-case g of 3.0% compresses implied price by \$162/share (\$482 → \$320).
3. The \$497 market price requires a terminal growth rate above 4.0% under any WACC scenario - confirming the overvaluation finding is robust.
4. The exit multiple cross-check at 20× implies \$574, above the entire GGM base-case WACC row, reflecting market pricing of AI optionality not captured in the GGM's normalised cash flow assumption.

Key Risks

Risk Factor	Bear Impact
CapEx remains elevated beyond FY2027	-\$40-80/share on GGM implied price
Azure growth decelerates faster than modelled	-\$30-60/share
Multiple compression (tech sector de-rating)	-\$50-100/share on exit multiple
Regulatory / antitrust action (OpenAI, bundling)	Qualitative; unpredictable
USD strength (~50% international revenue)	-1-2% revenue per 5% USD move

Post-Valuation Update (May 2026)

By the analysis date, MSFT had declined from \$497 to approximately \$450 - moving ~10% toward the GGM base-case estimate of \$385. This directional movement is consistent with the model's overvaluation finding and partially validates the thesis. The GGM implied price remains below the current market price, and the overvaluation verdict is maintained.

Project Overview

Item	Detail
Objective	Build an institutional-grade DCF to estimate MSFT intrinsic value anchored to the FY2025 10-K
Deliverable	13-tab Excel model: 3-statement historical (FY2021-25), 5Y forecast, WACC, FCFF build, EV bridge, scenario toggle, sensitivity, executive summary
Data Sources	SEC EDGAR 10-K (FY2021-25), Damodaran implied ERP, FRED 10Y Treasury, Yahoo Finance / Bloomberg, MSFT long-term bond yields
Valuation Date	30 June 2025 - all balance sheet inputs and market price anchored to this date
Scenario Toggle	Single driver switch (Bear / Base / Bull) cascades automatically through all 13 tabs

Revenue Architecture - Three-Segment Build

Revenue is forecast at the segment level with explicit growth rationale anchored to the most recent 10-K commentary and observable trends.

Segment	FY2025 Revenue	Growth Logic (FY26-30)	FY2030E
Productivity & Business Processes	\$120.8B	12.5% → 8.0% - Copilot ramp, M365 SMB penetration	\$195.4B
Intelligent Cloud	\$106.3B	19.0% → 11.0% - Azure AI workloads, hyperscaler share	\$213.4B
More Personal Computing	\$54.6B	6.0% → 4.0% - Bing/Copilot offsets PC hardware decline	\$69.1B
Total	\$281.7B	~11.1% CAGR	\$477.8B

FCFF Forecast

Metric	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Total Revenue (\$B)	320.3	359.6	400.0	439.6	477.8
EBIT Margin	46.0%	46.0%	46.0%	46.0%	46.0%
CapEx % Revenue	22.0%	22.0%	18.0%	15.0%	12.5%
FCFF (\$B)	83.1	94.0	121.1	146.9	172.3
FCF Margin	25.9%	26.1%	30.3%	33.4%	36.1%

CapEx normalisation is the single highest-impact assumption in the model. By FY2030, CapEx at 12.5% of revenue is consistent with Microsoft's pre-AI historical baseline (FY2022: 12.0%, FY2023: 13.3%), driving FCF margin recovery from 25.9% to 36.1%.

WACC Derivation

Input	Value	Source / Rationale
Risk-Free Rate (Rf)	4.42%	10Y US Treasury yield - FRED, as of valuation date
Equity Risk Premium (ERP)	4.18%	Damodaran implied ERP - updated annually
Beta (β)	0.90	5Y weekly regression vs. S&P 500, re-levered; adjusted for net cash balance sheet

Cost of Equity (Ke)	8.18%	CAPM: 4.42% + 0.90 × 4.18%
Pre-tax Cost of Debt (Kd)	4.77%	YTM on MSFT long-term bond yields
After-tax Cost of Debt	3.91%	$K_d \times (1 - 18\% \text{ effective tax rate})$
Equity Weight	98.55%	Market-value weights per CFA/IB convention
Debt Weight	1.45%	Total Debt / (Market Cap + Total Debt)
WACC	8.12%	Cross-check vs. Assumptions tab: consistent ✓

Key Analytical Insights

CapEx cycle is the valuation swing factor. The difference between bear (\$222) and bull (\$494) GGM implied prices is driven almost entirely by CapEx assumptions. At 22% through FY2030, FCF margin stays suppressed at ~26%. At 12.5% (base), FCF margin recovers to 36%+, consistent with Microsoft's pre-AI cash generation profile.

The market requires an unrealistic terminal growth assumption to justify \$497. Justifying the current market price requires g above 4.0% under the plausible assumed WACC for the base case - well above long-run US nominal GDP growth, leaving no margin for error.

Exit multiple is structurally above GGM in this environment. The \$189 gap between \$574 (exit multiple) and \$385 (GGM) represents the market's implicit premium for AI optionality - the portion of current pricing that cannot be explained by normalised long-run cash flows.

Post-valuation price action directionally validates the model. The ~10% decline from \$497 to ~\$450 between June 2025 and May 2026 is consistent with the overvaluation finding. The GGM verdict of -23% implied more correction than has occurred, but the directional signal was correct.

Skills Demonstrated

✓ Full integrated 3-statement model built from 10-K source data (FY2021-FY2025)
✓ Segment-level revenue forecasting with explicit, documented growth rationale
✓ WACC derivation using market-value weights, Damodaran ERP, and re-levered beta
✓ FCFF waterfall with correct sign conventions, NWC schedule, and CapEx roll-forward
✓ Dual terminal value methods (GGM primary; exit multiple as cross-check)
✓ Single-driver scenario toggle (Bear / Base / Bull) cascading through all 13 tabs
✓ 2D sensitivity analysis across $WACC \times g$ and $WACC \times EV/EBITDA$ multiple
✓ Investment memo writing calibrated for sophisticated financial audiences
✓ PP&E roll-forward and CapEx normalisation modelling with documented rationale (<i>simplified D&A</i>)

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